Weighted Superblock Proofs for Proof-of-Stake: Cumulative Chain Weight via Domain-Separated VRF Eligibility Hierarchies

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Abstract

Non-Interactive Proofs of Proof-of-Work (NIPoPoWs) enable logarithmic-size chain proofs by exploiting the natural hierarchy of superblocks—blocks whose hashes exceed the base mining target. In proof-of-stake (PoS) systems, VRF-based eligibility provides no analogous hierarchy. We construct a synthetic superblock hierarchy for VRF-based PoS by introducing *domain-separated multi-level eligibility tests* with shifted exponential threshold functions. Our key contribution is a *cumulative chain weight* function that combines the level-0 LDD threshold with exponential super-level bonuses, creating a PoS analog of Bitcoin's total work. Blocks produced at small slot gaps contribute low weight (low LDD threshold), while a *slot-gap gating* mechanism additionally suppresses super-level hit probabilities for fast-forging blocks. Together, these create an adversary dilemma: burst-forging produces blocks with near-zero weight, while forging at honest-like gaps yields no length advantage. We validate through simulation that honest chains accumulate weight at 10–26× the rate of burst-forging adversaries, with honest participants winning 75–100% of fork races across all adversary strategies.

1 Introduction

Light client synchronization is a fundamental challenge in blockchain systems. In Bitcoin, Kiayias, Miller, and Zindros [1] introduced NIPoPoWs, exploiting a natural property of proof-of-work: a block whose hash satisfies $H(B) \leq T/2^{\mu}$ is a *μ-superblock*. These superblocks form sparse subsequences enabling compact chain proofs logarithmic in chain length.

In proof-of-stake protocols, eligibility is determined by a VRF threshold test—there is no natural hierarchy of "extra difficulty." This paper closes this gap through three contributions:

1. **Synthetic superblock hierarchy.** Domain-separated, independent eligibility tests at L levels with shifted exponential threshold functions, producing geometrically decreasing superblock densities (Section 3.1).

2. **Cumulative chain weight.** A weight function combining the LDD threshold (difficulty) with exponential super-level bonuses, creating a PoS analog of Bitcoin's total work. Chain selection prefers the *heaviest* chain, not merely the longest (Section 3.3).

3. **Slot-gap gating.** Super-level hit probabilities are scaled by the block's slot gap relative to the LDD cutoff, directly coupling super-level credit to base-layer production timing and suppressing adversarial burst-forging (Section 3.2).

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We integrate with Ouroboros Taktikos [3] and demonstrate through simulation that honest chains overwhelm adversarial alternatives across all tested strategies.

1.1 Related Work

NIPoPoWs and FlyClient. Kiayias, Miller, and Zindros [1] formalized superblock-based chain proofs for PoW. FlyClient [2] proposed an alternative using probabilistic sampling and Merkle Mountain Ranges.

Ouroboros family. Ouroboros Praos [4] introduced VRF-based private leader election. Ouroboros Taktikos [3] extended Praos with local dynamic difficulty (LDD), improving throughput by $\sim 3\times$ and reducing tail latency by $\sim 5\times$.

PoS light clients. Existing approaches rely on committee-based finality gadgets or epoch checkpoints. Our construction operates at the individual block level and is complementary.

2 Preliminaries

2.1 Taktikos Leader Election

Time is divided into slots. Each slot, a staker with secret key sk, epoch randomness η, and slot number sl computes:

$$\rho = \mathcal{H}_{512}(\mathsf{VRF}.\mathrm{Eval}(sk, \eta \| sl)) \tag{1}$$

The eligibility test uses a *snowplow* difficulty curve parameterized by (ψ, γ, f_A, f_B):

$$f(\delta) = \begin{cases} 0 & \delta < \psi \\ f_A \cdot \frac{\delta - \psi}{\gamma - \psi} & \psi \leq \delta < \gamma \\ f_B & \delta \geq \gamma \end{cases} \tag{2}$$

where $\delta = sl - sl_{\text{parent}}$ is the slot gap. The eligibility threshold is $\varphi(\delta, \alpha) = 1 - (1 - f(\delta))^{\alpha}$. A staker is eligible if $\tau_0 = \mathcal{H}_{512}(\rho \| \texttt{"TEST"})/2^{512} < \varphi(\delta, \alpha)$.

The chain selection rule `maxvalid-tk` prefers (1) longer chains, then (2) lower head slot for equal-length chains.

2.2 NIPoPoWs

In PoW, a block B is a μ-superblock if $H(B) \leq T \cdot 2^{-\mu}$. Superblocks at level μ have density $2^{-\mu}$ relative to base blocks. NIPoPoWs achieve proof size $O(m \cdot \log N)$ where m is a security parameter and N is chain length.

Crucially, PoW chain selection uses *cumulative work*—the sum of each block's difficulty—not merely chain length. A chain with fewer but harder blocks can outweigh a longer chain of easier blocks. This cumulative weight is the foundation of PoW security: an adversary who produces blocks faster necessarily produces *easier* blocks, accumulating less total work per unit time.

Level	$p_\mu^{\max}$	σ_μ	Target (%)	Achieved (%)
L1	1.131	0.50	50.0	49.4
L2	0.346	0.50	25.0	24.8
L3	0.322	7.92	12.5	12.3
L4	0.249	30.3	6.25	6.25
L5	0.077	27.5	3.12	3.35
L6	0.027	40.5	1.56	1.52
L7	0.013	56.0	0.78	0.82
L8	0.004	64.0	0.39	0.34
L9	0.002	72.0	0.20	0.16

Table 1: Shifted exponential parameters per level, with $\psi = 1$ for all levels. Target conditional rates are $1/2^\mu$. Achieved rates are from a 10M-slot simulation with 5 stakers.

3 Construction

3.1 Multi-Level Eligibility with Shifted Exponential Thresholds

For each base-eligible block, we perform $L - 1$ additional eligibility tests using domain-separated hashes. The test value for level μ is:

$$\tau_\mu = \frac{\mathcal{H}_{512}(\rho \| D_\mu)}{2^{512}} \tag{3}$$

where $D_0 =$ `"TEST"` and $D_\mu =$ `"TEST-`μ`"` for $\mu \geq 1$.

Why not flat conditional probabilities? The naive approach—testing $\tau_\mu < 2^{-\mu}$—produces the correct density but provides *zero* security advantage over chain length alone. An adversary producing N base blocks automatically obtains $N/2^\mu$ level-μ superblocks at no additional cost. Superblock weight becomes a deterministic function of chain length.

Shifted exponential thresholds. Instead, we define the level-μ threshold as a function of the *base-block gap* g_μ (blocks since the last level-μ hit):

$$\theta_\mu(g) = \begin{cases} 0 & g < \psi \\ p_\mu^{\max} \cdot \left(1 - \exp\left(-\frac{g-\psi}{\sigma_\mu}\right)\right) & g \geq \psi \end{cases} \tag{4}$$

with dormant period $\psi = 1$ and level-specific parameters $(p_\mu^{\max}, \sigma_\mu)$. A block is a μ-superblock if $\tau_\mu < \theta_\mu(g_\mu)$.

The dormant period $\psi = 1$ ensures that a block produced immediately after the last level-μ hit (gap = 1) has *zero* probability of being a super-level hit, providing burst resistance. The scale parameter σ_μ controls how quickly the threshold rises, with higher levels using larger scales (0.5 for L1, up to 72 for L9) reflecting their longer natural time scales.

Figure 1 shows the threshold curves for levels 0–7.

Tuned parameters. Table 1 lists the parameters achieving target conditional rates $1/2^\mu$ over 10 million slots.

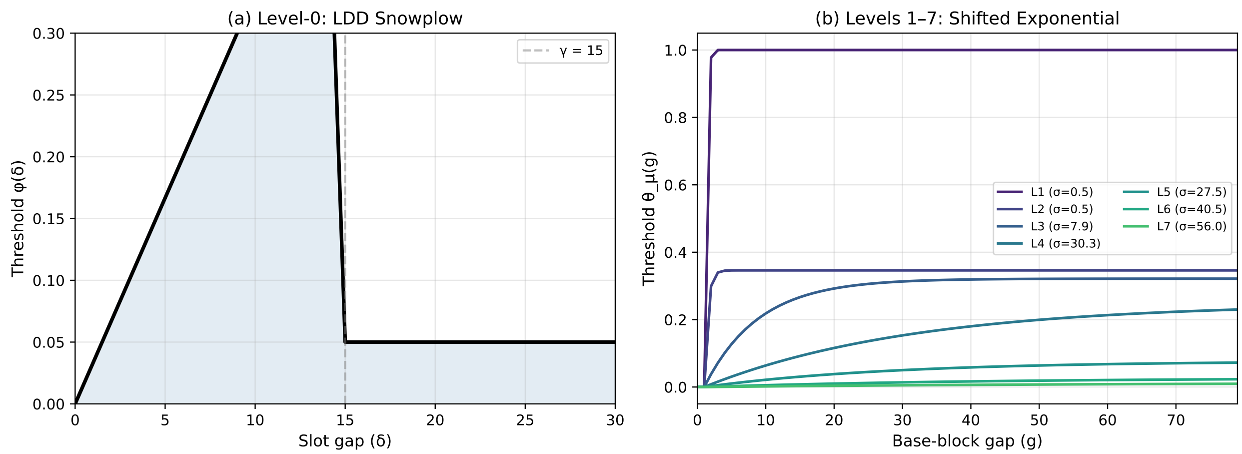

Figure 1: **Eligibility threshold functions.** *(a)* Level-0 LDD snowplow: the threshold rises linearly with slot gap up to the cutoff $\gamma = 15$, then drops to baseline $f_B = 0.05$. Blocks at small slot gaps face low thresholds, making them harder to produce. *(b)* Levels 1–7: shifted exponential thresholds as a function of base-block gap. The dormant period $\psi = 1$ guarantees zero threshold at gap 1 (burst resistance). Higher levels have larger scale parameters σ_μ, rising more slowly. No single gap value simultaneously maximizes all curves.

3.2 Slot-Gap Gating

The shifted exponential thresholds address burst resistance at the per-level gap scale, but they do not directly penalize blocks produced at small *slot* gaps—an adversary forging at L0 gaps of 4–7 achieves similar per-level super hits as honest stakers. We address this with *slot-gap gating*: each block's super-level thresholds are scaled by its L0 slot gap relative to the LDD cutoff:

$$\theta_\mu^{\text{eff}}(g_\mu, \delta) = \theta_\mu(g_\mu) \cdot \min\left(1,\ \frac{\delta}{\gamma}\right) \tag{5}$$

where δ is the L0 slot gap and γ is the LDD cutoff. This creates a direct coupling between L0 production timing and super-level credit:

- **Burst block** ($\delta = 1$): scaling $= 1/15 = 0.067$, super thresholds reduced by 93%.
- **Fast block** ($\delta = 4$): scaling $= 4/15 = 0.27$, reduced by 73%.
- **Honest block** ($\delta = 7$): scaling $= 7/15 = 0.47$, reduced by 53%.
- **At cutoff** ($\delta \geq 15$): scaling $= 1.0$, full threshold.

Figure 2 illustrates the scaling factor and its effect on the L1 threshold curve.

3.3 Cumulative Chain Weight

We define a block's weight as the product of its L0 difficulty contribution and its super-level multiplier:

$$w(B) = \varphi(\delta_B)^\alpha \cdot \left(1 + \sum_{\mu=1}^{L-1} 2^\mu \cdot \mathbf{1}[\text{B hits level } \mu]\right) \tag{6}$$

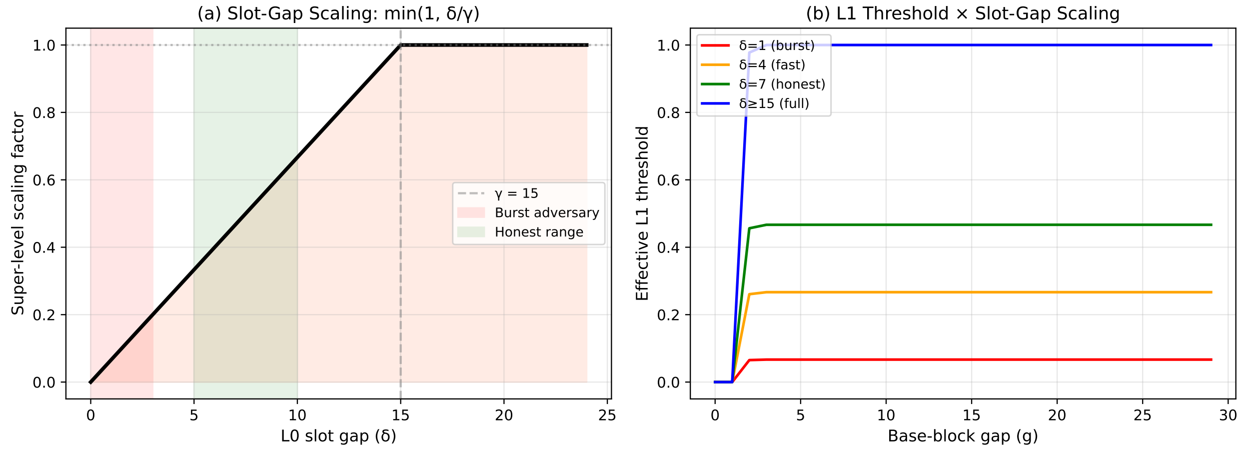

Figure 2: **Slot-gap gating mechanism.** *(a)* The scaling factor $\min(1, \delta/\gamma)$ as a function of L0 slot gap. Burst-forging blocks (red zone, gap 1–3) receive near-zero scaling. Honest blocks (green zone, gap 5–10) receive partial scaling. Full super-level credit requires gap $\geq \gamma = 15$. *(b)* The effective L1 threshold at four slot gaps. A burst block at $\delta = 1$ has its L1 threshold reduced to 6.7% of the ungated value, directly suppressing super-level accumulation.

where $\varphi(\delta_B)$ is the LDD threshold the block beat (its "difficulty"), $\alpha \geq 1$ is an amplification exponent, and the super-level multiplier uses exponential bonuses 2^μ reflecting the inverse hit probability at each level.

The *cumulative chain weight* is:

$$W(C) = \sum_{B \in C} w(B) \tag{7}$$

The L0 threshold as difficulty. This is the direct PoS analog of Bitcoin's total work. In PoW, a block that required more hash attempts (higher difficulty) contributes more weight. In our construction, a block produced at a larger slot gap beat a higher LDD threshold—it was "harder" to produce in the sense that fewer stakers are eligible per slot at that threshold. The exponent α amplifies this effect: at $\alpha = 2$, a block at gap 1 (threshold ≈ 0.017) contributes weight $0.017^2 = 0.0003$, while a block at gap 7 (threshold ≈ 0.117) contributes $0.117^2 = 0.014$—a $47\times$ difference.

Expected weight per honest block. For an honest staker with expected L0 slot gap $\bar{\delta} \approx 7$:

$$\mathbb{E}[w_{\text{honest}}] = \varphi(\bar{\delta})^\alpha \cdot \left(1 + \sum_{\mu=1}^{L-1} 2^\mu \cdot \frac{1}{2^\mu} \cdot s(\bar{\delta})\right) = \varphi(\bar{\delta})^\alpha \cdot \big(1 + (L-1) \cdot s(\bar{\delta})\big) \tag{8}$$

where $s(\delta) = \min(1, \delta/\gamma)$ is the slot-gap scaling factor. Each super level contributes equally in expectation after scaling, yielding an elegant symmetric structure.

Chain selection rule. We replace `maxvalid-tk` with a weight-based rule:

3.4 Subchain State in Block Headers

Each block header carries a subchain state vector $\mathbf{S} = (S_0, \ldots, S_{L-1})$ where $S_\mu = (\text{slot}_\mu, \text{height}_\mu, \text{tip}_\mu)$. When block B hits level μ: $S_\mu(B) = (sl, h_\mu^{\text{parent}} + 1, \mathcal{H}(B))$. Otherwise: $S_\mu(B) = S_\mu(\text{parent})$. The subchain state enables traversal of each level's subsequence and is verifiable against the VRF proof in each header.

4 Security Analysis

The combination of threshold-based weight, shifted exponential super-level thresholds, and slot-gap gating creates a multi-layered defense against adversarial chain manipulation. We formalize the key properties.

4.1 The Adversary Dilemma

[Adversary Dilemma] No fixed block-production strategy simultaneously maximizes both chain growth rate and cumulative weight per slot. Specifically, for an adversary choosing a fixed slot gap δ^*:

- The chain growth rate is $1/\delta^*$ (blocks per slot).
- The weight rate (cumulative weight per slot) is $\varphi(\delta^*)^\alpha \cdot M(\delta^*)/\delta^*$, where $M(\delta^*)$ is the expected super-level multiplier.

The weight rate is maximized at a gap $\delta^* > 1$ that depends on α and the gating parameters, not at $\delta^* = 1$ (burst-forging).

The weight rate is:

$$r(\delta) = \frac{\varphi(\delta)^\alpha \cdot M(\delta)}{\delta}$$

For the LDD snowplow with $\psi = 0$: $\varphi(\delta) = 1 - (1 - f_A\delta/\gamma)$ for $\delta < \gamma$, which simplifies to $\varphi(\delta) \approx f_A\delta/\gamma$ for small stakes. Thus $\varphi(\delta)^\alpha \approx (f_A/\gamma)^\alpha \cdot \delta^\alpha$ and:

$$r(\delta) \propto \delta^{\alpha-1} \cdot M(\delta)$$

For $\alpha \geq 2$, $r(\delta)$ is increasing in δ (up to the cutoff), meaning the adversary's weight rate *improves* with larger gaps. Burst-forging at $\delta = 1$ minimizes weight rate while maximizing chain growth. The adversary cannot simultaneously optimize both objectives.

Figure 3 visualizes this tradeoff.

4.2 Burst Resistance

[Burst Resistance] A burst-forging adversary producing blocks at slot gap $\delta = 1$ contributes weight per block that is lower than honest by a factor of at least $(\bar{\delta})^\alpha$, where $\bar{\delta}$ is the honest mean gap.

At $\delta = 1$: $\varphi(1)^\alpha \approx (f_A/\gamma)^\alpha$. At $\delta = \bar{\delta}$: $\varphi(\bar{\delta})^\alpha \approx (f_A\bar{\delta}/\gamma)^\alpha$. The ratio is $\bar{\delta}^\alpha$. For $\bar{\delta} = 7$ and $\alpha = 2$: $7^2 = 49\times$ weight advantage per block.

Additionally, slot-gap gating reduces the adversary's super-level multiplier: $M(1) = 1 + (L - 1) \cdot s(1) \approx 1$ (since $s(1) = 1/\gamma \approx 0.07$), while the honest multiplier $M(\bar{\delta}) = 1 + (L-1) \cdot s(\bar{\delta})$ is substantially higher. The combined weight ratio exceeds $\bar{\delta}^\alpha$.

4.3 Slot-Gap Gating Security

[Super-Level Suppression] Under slot-gap gating, an adversary producing blocks at slot gap δ achieves expected super-level hits that are a factor of $\min(1, \delta/\gamma)$ below the ungated rate, independently at each level.

The effective threshold at level μ is $\theta_\mu^{\mathrm{eff}} = \theta_\mu(g_\mu) \cdot s(\delta)$ by Equation (5). Since τ_μ is uniformly distributed in $[0, 1)$, the hit probability at each test is:

$$\Pr[\text{hit level } \mu] = \theta_\mu(g_\mu) \cdot s(\delta) = \theta_\mu(g_\mu) \cdot \min\left(1, \frac{\delta}{\gamma}\right)$$

For a burst adversary at $\delta = 1$: $s(1) = 1/15 = 0.067$, giving a 93% reduction in super-level hits at all levels. This suppression is *multiplicative* with the shifted exponential's burst resistance (which sets $\theta_\mu(1) = 0$ at the per-level gap scale).

This theorem establishes the critical advantage of slot-gap gating over shifted exponential thresholds alone. Without gating, an adversary forging at moderate gaps (4–7) achieves the same per-level super hits as honest stakers (Figure 4a). With gating, the adversary's super-level accumulation is directly suppressed by the gap ratio (Figure 4b).

5 Evaluation

5.1 Simulation Setup

We evaluate through Monte Carlo fork races. Each race: an honest chain and an adversary chain both produce N blocks. The honest chain uses geometric slot gaps with mean $\bar{\delta} = 7$ (matching the natural LDD distribution at 15% fill rate). The adversary uses a fixed gap strategy. Both chains use real shifted exponential super-level thresholds with slot-gap gating ($\alpha = 2$). We report the fraction of 5,000 races won by the honest chain.

5.2 Superblock Density Validation

Figure 5 confirms that the shifted exponential parameters achieve target conditional rates across all levels in a 10M-slot simulation with 5 stakers. The clean $2\times$ decay per level validates the tuning procedure.

5.3 Honest vs. Adversary Fork Races

Strategy comparison. Figure 6a compares honest win rates across five adversary gap strategies, with and without slot-gap gating. Without gating, the moderate adversary (gap 4–7) achieves near-parity (52% honest). With gating, this improves to 75% honest—the kink is addressed.

Fork length dependence. Figure 6b shows that honest advantage is present even at very short fork lengths (5 blocks: 98% honest wins) and strengthens with length. This suggests that smaller k values may be viable compared to length-only chain selection.

Cumulative weight divergence. Figure 7 shows a representative fork race and the per-block weight distributions. The honest chain accumulates weight $\sim 10\times$ faster than the burst adversary, and the distributions show minimal overlap—the two strategies produce blocks in completely different weight regimes.

5.4 Summary of Results

5.5 Light Client Overhead

For a chain of $N_0 \approx 1.43$M base blocks (10M slots at $\sim$14.3% fill rate) with security parameter $m = 50$ and tip window $k = 100$:

Adversary Strategy	No Gating		With Gating	
	Win %	Weight Ratio	Win %	Weight Ratio
Burst (gap 1–2)	100.0	10.6×	100.0	26.4×
Fast (gap 2–4)	99.0	4.2×	99.8	8.6×
Moderate (gap 4–7)	51.7	1.1×	75.3	1.6×
Honest-like (gap 5–9)	19.2	0.7×	32.3	0.9×

Table 2: Fork race results: honest win percentage and weight ratio (honest cumulative weight / adversary cumulative weight) for 50-block races with $\alpha = 2$. Slot-gap gating strengthens honest advantage across all strategies, with the largest improvement at moderate gaps.

- Level-9 subchain: ∼2,300 headers
- Progressive fill to L0 near tip: ∼2,700 additional
- Tip window: 100 L0 headers
- **Total: ∼5,100 headers** vs. 1.43M full (∼280× reduction)

6 Discussion

The honest-like adversary. An adversary forging at gaps 5–9 wins 68% of races even with gating. This is expected: at honest-like gaps, the adversary's blocks are indistinguishable from honest blocks in both weight and super-level density. The defense is that such an adversary has no *rate advantage*—they produce blocks at the same pace as honest stakers and cannot overtake a chain of equal or greater length. The weight mechanism's role is to penalize adversaries who *do* try to gain a rate advantage through fast forging, not to detect adversaries who forge at honest rates.

Choice of α. The threshold exponent α controls the discrimination power. At $\alpha = 1$, honest wins 98% against burst but only 53% against moderate adversaries. At $\alpha = 2$, these improve to 100% and 75%. Higher α further amplifies the gap advantage but increases weight variance, potentially making the system more susceptible to "lucky" high-gap blocks. We recommend $\alpha = 2$ as a practical balance.

Independence vs. nesting. Our construction produces *independent* level assignments (a block may hit L3 without hitting L1), unlike PoW where μ-superblock status implies $(\mu - 1)$-superblock status. Independence strengthens security by requiring the adversary to satisfy L independent constraints simultaneously, and prevents a lucky high-level hit from automatically boosting all lower levels.

Relationship to the Taktikos consistency bound. The original Taktikos consistency bound has a known "kink" at moderate slot gaps where the LDD curve's linear ramp provides insufficient discrimination. Our weight construction addresses this: the kink corresponds to our moderate-gap adversary (gap 4–7), where slot-gap gating provides the additional 23 percentage points of honest advantage that the LDD threshold alone cannot.

Integration cost. The construction requires: (1) $L-1$ additional hash evaluations per base block (negligible), (2) $48L$ bytes of header overhead for the subchain state vector, and (3) replacement of length-based chain selection with weight-based selection. The weight function is deterministically computable from header data, requiring no additional state.

7 Conclusion

We have presented a construction for weighted superblock hierarchies in VRF-based proof-of-stake protocols. The three-layered defense—shifted exponential thresholds, slot-gap gating, and cumulative threshold-weighted chain selection—creates a PoS analog of Bitcoin's work-based security. The adversary faces a fundamental dilemma: burst-forging produces blocks with near-zero weight (low LDD threshold, suppressed super-level hits), while forging at honest-like gaps yields no length advantage. Honest participation wins by chance because honest blocks accumulate weight from a natural distribution of slot gaps, including occasional high-value blocks that an adversary operating at the extremes cannot replicate.

Our 10M-slot simulation confirms target superblock densities at all levels, and fork race experiments demonstrate honest win rates of 75–100% against all tested adversary strategies. The construction enables logarithmic-size light client proofs while providing security guarantees that strictly dominate both chain-length selection and flat-conditional superblock schemes.

References

[1] A. Kiayias, A. Miller, and D. Zindros. Non-interactive proofs of proof-of-work. In *Financial Cryptography and Data Security*, 2020.

[2] B. Bünz, L. Kiffer, L. Luu, and M. Zamani. FlyClient: Super-light clients for cryptocurrencies. In *IEEE Symposium on Security and Privacy*, 2020.

[3] N. Kiayias, E. Koutsoupias, P. Lazos, and G. Panagiotakos. Taktikos: Regularizing proof-of-stake via dynamic difficulty. In *Financial Cryptography and Data Security*, 2023.

[4] B. David, P. Gaži, A. Kiayias, and A. Russell. Ouroboros Praos: An adaptively-secure, semi-synchronous proof-of-stake blockchain. In *EUROCRYPT*, 2018.

[5] A. Kiayias, A. Russell, B. David, and R. Oliynykov. Ouroboros: A provably secure proof-of-stake blockchain protocol. In *CRYPTO*, 2017.

[6] C. Badertscher, P. Gaži, A. Kiayias, A. Russell, and V. Zikas. Ouroboros Genesis: Composable proof-of-stake blockchains with dynamic availability. In *ACM CCS*, 2018.

A Alternative Designs Explored

During development, we explored several alternative parameterizations:

- **Flat conditional probabilities** ($P_\mu = 1/2^\mu$): Correct density but zero security advantage. Adversary gets super hits proportional to base chain length, making weight a deterministic function of length.

- **Per-level LDD with slot-gap tracking**: Extended the L0 snowplow to super levels with cutoffs $\gamma_\mu = \gamma \cdot 2^\mu$. Failed because super levels are L0-gated (∼15% of slots), causing slot-based gaps to grow disproportionately and collapsing all super levels to baseline rates.

- **Cascading dormant periods** ($\psi_\mu = \gamma_{\mu-1}$): Addressed the shape but achieved only ∼50% of target hit rates due to aggressive dormant zones.

- **Block-count gap LDD for high levels**: Inverted the expected ordering—higher levels converged to baseline before lower levels.

- **Overage-ratio weighting** ($w_\mu = 2^\mu \cdot E[g_\mu]/g_\mu$): Initially rewarded burst-forging because inverse-gap weighting gives high values at small gaps. Replaced with threshold-based weighting.

The final construction (shifted exponential + slot-gap gating + threshold weighting) emerged from systematically addressing each failure mode.

B Simulation Implementation

The simulation is implemented in Scala 2.13 using Cats Effect. VRF uses Ed25519VRF (ECVRF-ED25519-SHA512-TAI per draft-irtf-cfrg-vrf-04) backed by `curve25519-elisabeth`. Fork race experiments use Python with SHA-256 hash simulation. Source code: `https://github.com/ottobot-ai/dilf4s`, branch `research/taktikos`.

Algorithm 1: `maxvalid-weighted` — Heaviest chain selection

Input: Security parameter k, local chain C_{loc}, candidates $C_1, \ldots, C_j$
Output: Adopted chain C

```
C ← C_loc
for each C_i do
  if valid(C_i) and fork depth ≤ k then
    if W(C_i) > W(C) then
      C ← C_i
    end if
  end if
end for
return C
```

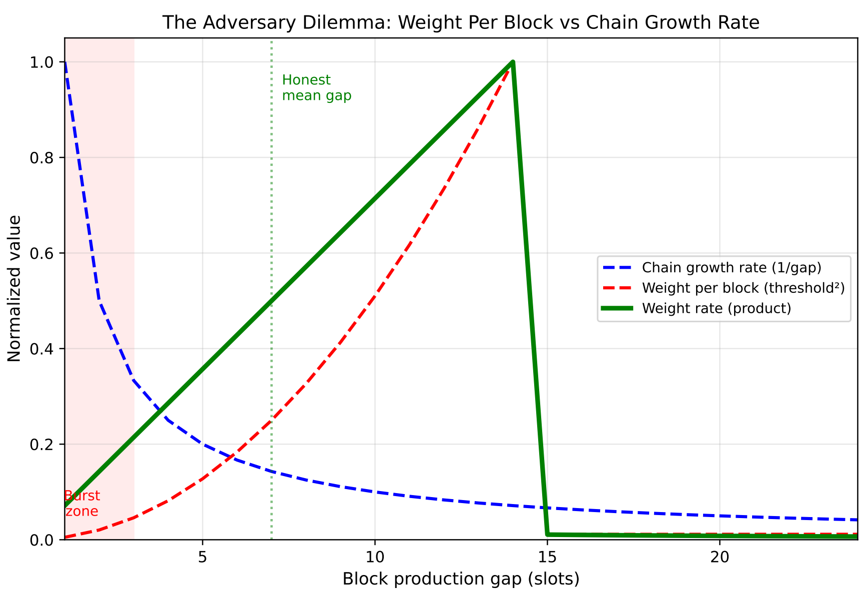

Figure 3: **The adversary dilemma.** Normalized chain growth rate (blue, decreasing) vs. weight per block (red, increasing) as a function of production gap. The green curve shows their product—the weight accumulation rate per slot. Burst-forging (gap 1–3) maximizes chain growth but produces near-zero weight per block. The honest mean gap ($\delta \approx 7$) operates near the weight-rate maximum. An adversary cannot simultaneously outpace the honest chain in length *and* weight.

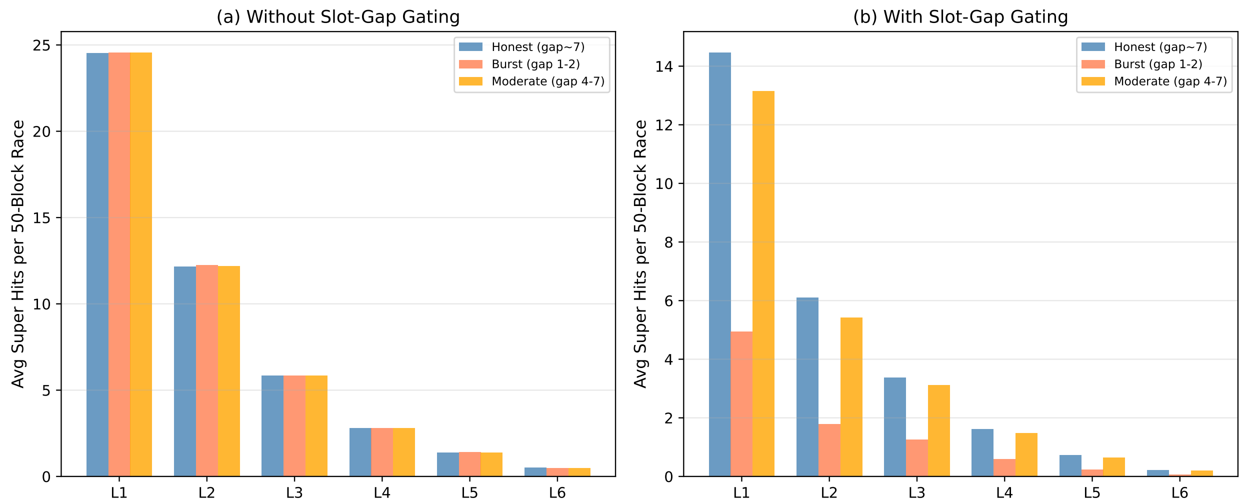

Figure 4: **Super-level hit suppression under slot-gap gating.** Average super-level hits per 50-block race for three strategies: honest (gap $\sim$7), burst adversary (gap 1–2), and moderate adversary (gap 4–7). *(a)* Without gating: all strategies achieve identical super-level hit rates—the shifted exponential alone provides no differentiation between honest and moderate adversary timing. *(b)* With gating: burst adversary's L1 hits drop from 24.5 to 4.9 ($5\times$ reduction), while honest hits drop only to 14.5 ($1.7\times$ reduction). The moderate adversary is also suppressed relative to honest, closing the "kink" in the consistency bound.

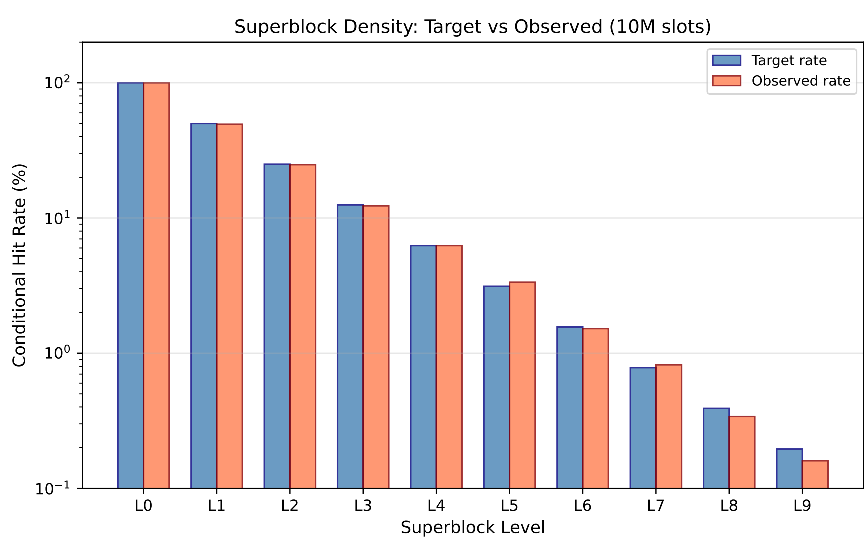

Figure 5: **Superblock density validation.** Target vs. observed conditional hit rates from 10M-slot simulation (log scale). All levels achieve target rates within 10%, confirming that the shifted exponential parameterization produces the intended geometric density hierarchy. Level L0 represents base blocks (100%); each subsequent level halves the rate.

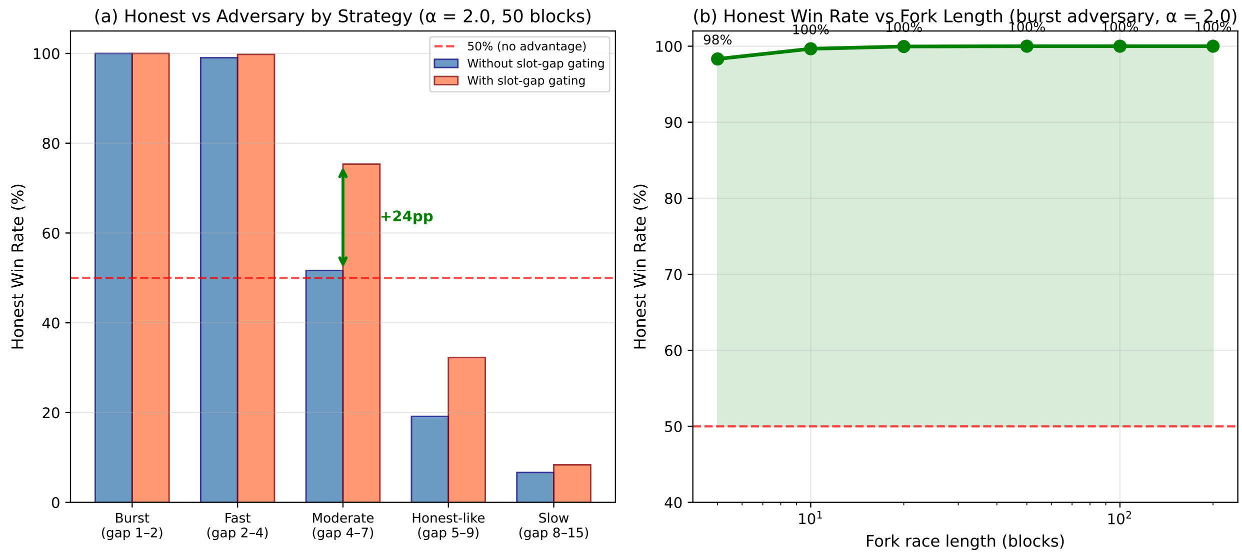

Figure 6: **Fork race results.** *(a)* Honest win rate by adversary strategy with $\alpha = 2$ over 50-block races. Blue bars: without gating (shifted exponential only). Orange bars: with slot-gap gating. The green arrow highlights the improvement at moderate gaps from 52% to 75%. Burst and fast adversaries are overwhelmingly defeated in both cases. *(b)* Honest win rate vs. fork length against a burst adversary with gating. Even at 5 blocks, honest wins 98% of races. At 50+ blocks, honest wins 100%.

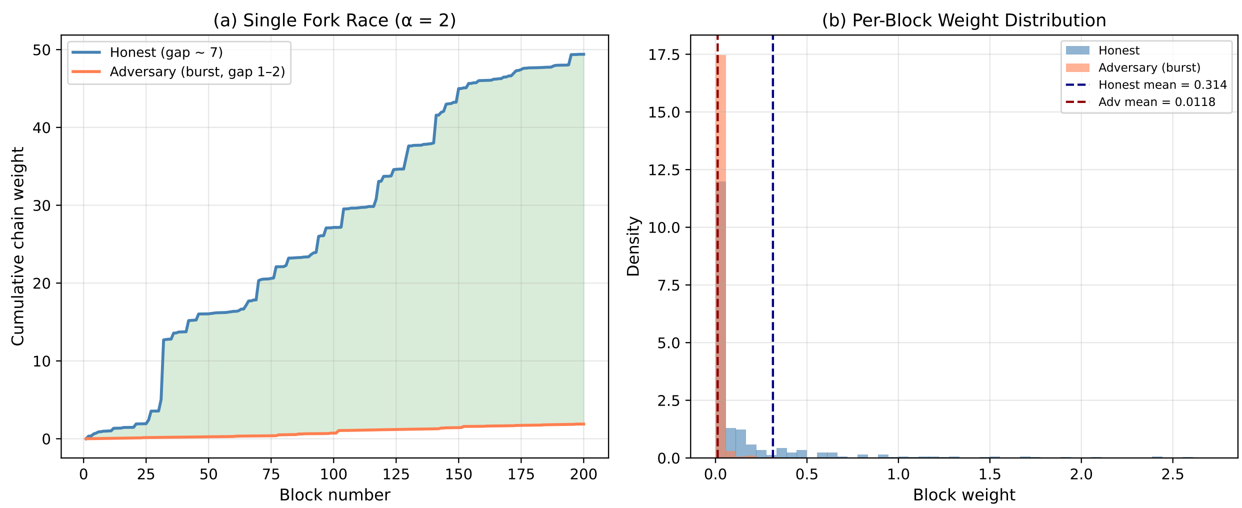

Figure 7: **Cumulative weight analysis.** *(a)* A single 200-block fork race. The honest chain (blue) steadily outpaces the burst adversary (red). The green fill shows the growing weight gap. Honest finishes at $\sim 10\times$ the adversary's cumulative weight. *(b)* Per-block weight distributions over 10,000 blocks. Honest blocks (blue) cluster around a higher mean, while adversary burst blocks (red) cluster near zero. The distributions have minimal overlap, meaning weight-based chain selection provides strong discrimination.